

Certified Topological Pruning for Spacecraft-Component Perception: P0–P1 Progress Report

Certified Topological Pruning workstream
topo_prune / seg_masks subtrees, PEEK-TiNA repository

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Abstract

The parent project ranks convolutional channels by a persistence-derived score, prunes the low scorers, and reports that accuracy holds. That is a *correlational* claim. This report documents the first two phases (P0, P1) of a plan to replace it with an *interventional and certified* one. P0 (discovery) established that the target model is a YOLO detector, not the segmentation network the plan assumed; we retarget the certified object to the detector’s per-scale, per-class classification *logit maps*, whose superlevel sets carry genuine homology. P1 builds the certificate — a per-image numerical check, backed by the Cohen–Steiner–Edelsbrunner–Harer bottleneck stability theorem, that predicts *before any retraining* which pruned inputs provably preserve the Betti numbers of the thresholded prediction — and its falsification audit. We report two experiments run on this workstation. E1 prunes the classification head of a COCO-pretrained YOLO26-nano at ten budgets and audits 5,550 (image, class, scale) triples: the sound certified bound produces **zero audited violations** on every arm, at a certified coverage that falls from 27.9 % at a 10 % head-channel budget to near zero by 40 %. E2 validates the certificate against 240 synthetic scenes with exact ground-truth homology: **zero constructed-versus-computed mismatches and zero audit violations** across 8,400 perturbed cases, with certified coverage decaying smoothly as the perturbation grows past each scene’s margin. A parallel branch derives segmentation masks from bounding boxes (GrabCut) and measures, on a COCO stand-in, that the masks carry topology the boxes do not on 83 % of images. We close with an extended future-work section: a worst-case (rather than calibration-based) certified bound via interval bound propagation, wider pruning scope, the matched-FLOP two-sided intervention (P2), the Euler-characteristic control, and topological restoration (P4).

Contents

1	Introduction	2
1.1	The scientific problem	2
1.2	The phase structure	3
1.3	What P0 discovered: a model-class mismatch	3
2	The certificate	4
2.1	Objects	4
2.2	The stability bound	4
2.3	The two ε bounds	4
2.4	The certification check	5
2.5	The falsifiable prediction and its audit	5

3	Implementation	6
3.1	The retarget, precisely	6
3.2	Package	6
3.3	TDA backend and the Euler control	7
3.4	Pruning scope	7
4	Experimental setup	7
5	E1: falsification audit on a really-pruned YOLO26-nano	8
5.1	Findings	8
6	E2: synthetic validation with exact ground-truth homology	9
6.1	Findings	9
7	Option B: segmentation masks from bounding boxes	10
7.1	Method	10
7.2	Pipeline check on a COCO stand-in	10
8	Threats to validity	11
9	Future work	11
9.1	A worst-case certified bound via interval bound propagation	11
9.2	Widening the pruning scope	12
9.3	The certified pruning objective (P2/P3)	12
9.4	The two-sided intervention (P2)	12
9.5	The Euler-characteristic control	12
9.6	P0 hygiene, retrofitted	12
9.7	Topological restoration (P4)	13
9.8	Multi-scale certification strategy	13
9.9	Zigzag persistence as a stopping rule (P5, stretch)	13
9.10	Finishing Option B	13
9.11	On-device measurement (P3)	13
10	Conclusion	13
A	Reproduction	14
B	Key definitions and conventions	14

1 Introduction

1.1 The scientific problem

The parent repository (PEEK-TiNA) implements a topology-guided pruning criterion for a low-SWaP spacecraft-component perception model. The criterion computes a persistence-derived score for each channel, prunes the low scorers, retrains, and reports that detection accuracy is retained. Three objections apply to that as a scientific result, and a reviewer will raise all three:

- (i) **It may be a cheap statistic in disguise.** Neural persistence — an ICLR 2019 construction of the same family — was later shown to be approximately the *variance of the weights* in deep layers, with permuted weights barely moving the score. Until shown otherwise, the same must be assumed here.
- (ii) **No intervention.** Showing that pruning the low-score channels preserves accuracy does not show the score *caused* it. A FLOP-matched random removal might preserve accuracy equally well.
- (iii) **No guarantee.** A score is a ranking, not a statement about what the pruned model preserves. Rankings are heuristics; heuristics get benchmarked away.

The escape is not a better score. It is a claim with a truth value:

Given a pruning mask M and a perturbation bound $\varepsilon(M)$, we can certify — before any retraining — that the pruned model’s predicted segmentation has the same Betti numbers as the unpruned model’s, on every input where a computable margin condition holds. And we can falsify this: on any certified input where the Betti numbers in fact differ, the certificate is wrong.

That is a falsifiable prediction about a deployed model.

1.2 The phase structure

The plan proceeds $P0 \rightarrow P5$, each phase with an exit criterion and a kill criterion. This report covers:

P0 — Discovery and hygiene

Map the repository; reproduce the existing criterion; run the variance/permutation ablations that gate everything else.

P1 — Build and audit the certificate

Implement cubical persistent homology on the prediction, the margin test, the ε bounds, and the falsification harness; validate on synthetic data with known homology.

P2 (the two-sided intervention), P3 (budget sweep and on-device measurement), P4 (topological restoration), and P5 (mediation analysis) are future work (Section 9).

1.3 What P0 discovered: a model-class mismatch

The plan is written for a semantic *segmentation* network. Its central object is a per-class logit map $f_c : \Omega \rightarrow \mathbb{R}$ on the full pixel grid Ω ; the predicted mask is the superlevel set $S_c = \{f_c \geq \tau\}$; the falsifiable claim is about $\beta_k(S_c)$.

The repository trains **YOLOv5-nano** and **YOLO26-nano object detectors**. The detection head emits decoded boxes, not a logit map on the pixel grid. The existing topological criterion runs on *PEEK saliency maps* of internal activations — a construction squarely in the correlational quadrant the plan is trying to leave.

Rather than abandon the plan or change the model, we retarget the certified object (Section 3.1): the YOLO26 **Detect** head computes, for each detection level i and class c , a spatial classification logit map $f_c^{(i)} : \text{grid}_i \rightarrow \mathbb{R}$ *before* any sigmoid or box decode. Its superlevel sets $\{f_c^{(i)} \geq \tau\}$

have genuine homology, and every step of the plan’s certificate transfers with only interpretation changes. This is “option A” of the discovery memo. A parallel “option B” (Section 7) derives real segmentation masks from the boxes so the plan applies verbatim; its feasibility is still open.

2 The certificate

We implement the *check*, not the theorem: the code decides certification numerically, per image, per class, per scale; the theorem is the writeup around it.

2.1 Objects

Fix a decision threshold τ (the operating point of the deployed model, in logit units: for a confidence p , $\tau = \text{logit}(p) = \ln \frac{p}{1-p}$). For class c and detection level i :

- $f_c^{(i)} : \text{grid}_i \rightarrow \mathbb{R}$ — the classification logit map, hooked at the output of `one2one_cv3[i]` in the `Detect` head.
- $S_c^{(i)} = \{p : f_c^{(i)}(p) \geq \tau\}$ — the predicted high-confidence region for that class at that scale.
- $D(f_c^{(i)})$ — the **cubical persistence diagram** of the superlevel-set filtration of $f_c^{(i)}$. Computed with GUDHI’s `CubicalComplex` on $-f_c^{(i)}$ (GUDHI computes a sublevel/lower-star filtration; superlevel of f equals sublevel of $-f$), then mapped back to f -value units. In the returned diagram birth $\geq$ death; an essential class has death $= -\infty$. This is near-linear in pixels: there is no Vietoris–Rips complex anywhere in this project.
- $f_c^{(i),M}$ — the same logit map from the network pruned by mask M .

For an input of 640 px the three grids are 80×80 , 40×40 , 20×20 (strides 8/16/32). *Reading* β_k : a dimension- k feature with interval (b, d) is alive at τ iff $b \geq \tau > d$; the count of such features is exactly $\beta_k(S_c^{(i)})$.

2.2 The stability bound

By the Cohen–Steiner–Edelsbrunner–Harer stability theorem, for the sub/superlevel filtration of two functions on the same grid,

$$d_B\left(D(f_c^{(i)}), D(f_c^{(i),M})\right) \leq \|f_c^{(i)} - f_c^{(i),M}\|_\infty, \quad (1)$$

with Lipschitz constant exactly 1. Everything reduces to bounding the sup-norm of the logit perturbation caused by the mask.

2.3 The two ε bounds

We compute the right-hand side of Equation (1) two ways, and always report both and their ratio (the *looseness factor*).

Empirical, $\varepsilon_{\text{emp}}(M)$. The measured $\max_{x \in \text{calib}} \|f_c^{(i)}(x) - f_c^{(i),M}(x)\|_\infty$, per (c, i) . *Not a certificate*: it is a lower-bound reference measured on a finite set, and it can be exceeded on unseen inputs. It tells us how loose the analytic bound is.

Certified, $\varepsilon_{\text{cert}}(M)$. In this report the pruning mask removes input channels of the *final* 1×1 classification convolution W_{final} (see Section 3.4). That layer is linear, so a dropped input channel’s contribution to the logit map vanishes exactly, and the general Lipschitz product of the plan collapses to a closed form:

$$\varepsilon_{\text{cert}}(c, i) = \sum_{k \in \text{dropped}(i)} |W_{\text{final}}[c, i, k]| \cdot M_k, \quad M_k = \max_{x \in \text{calib}} \max_p |\phi_k^{(i)}(x)(p)|, \quad (2)$$

where $\phi^{(i)}$ is the penultimate head feature (the final conv’s input). M_k is a maximum over the calibration set, so Equation (2) is a *calibration-based high-confidence bound*, not a strict worst-case certificate; obtaining the latter requires an input-domain bound on $\phi^{(i)}$, i.e. interval bound propagation through the backbone, neck, and head prefix. That is the top P1 follow-up (Section 9.1).

2.4 The certification check

Definition 1 (Margin). Given a diagram D and threshold τ ,

$$m(\tau, D) = \min_{(b,d) \in D} \min(|b - \tau|, |d - \tau|),$$

the distance of the nearest birth or death to τ over all bars of all dimensions (essential bars contribute only their birth). An empty diagram gives $m = +\infty$.

Definition 2 (Certification). The triple (x, c, i) is **certified** for mask M if $m(\tau, D(f_c^{(i)}(x))) > \varepsilon(M)$ and $\ell_{\min} > 2\varepsilon(M)$, where $\ell_{\min}$ is the shortest lifetime among bars that straddle τ .

Bottleneck stability Equation (1) says every bar endpoint moves by at most ε . If no endpoint can cross τ under a move of that size, the set of bars alive at τ — and therefore $\beta_k(S_c^{(i)})$ for all k — is unchanged.

Proposition 1 (The second condition is implied by the first). *For our definition of m , $m(\tau, D) > \varepsilon \implies \ell_{\min} > 2\varepsilon$.*

Proof. For any bar (b, d) that straddles τ (so $b \geq \tau > d$),

$$b - d = (b - \tau) + (\tau - d) \geq 2 \min(b - \tau, \tau - d) \geq 2 m(\tau, D).$$

Hence $\ell_{\min} \geq 2 m(\tau, D) > 2\varepsilon$. □

We keep both conditions because the plan lists both and the redundant check is free; a reviewer preferring m to range only over non-straddling bars would make the second condition bite, so the redundancy is documented rather than removed.

2.5 The falsifiable prediction and its audit

On every certified triple, $\beta_k(S_c^{(i),M}) = \beta_k(S_c^{(i)})$ **for all k .**

`certificate.audit` checks this on every certified triple against the Betti numbers actually produced by the pruned model and emits a *violation count*. The expected value is zero.

- Violations = 0 across the test set $\implies$ the certificate is sound, and we have a topology-preservation guarantee for a pruned network.

Table 1: Segmentation certificate $\rightarrow$ detection certificate.

Plan (segmentation)	This report (detection)
one logit map f_c on the pixel grid	three logit maps $f_c^{(i)}$, one per detection level; certify each (c, i) pair independently, report per-level and all-levels-certified coverage
$\beta_0(S_c) = \text{mask components}$	$\beta_0(S_c^{(i)}) = \text{high-confidence blobs} \approx \text{predicted instance count}$; $\beta_1 = \text{enclosed holes}$ (a truss or a ring antenna is a real $\beta_1 = 1$)
stability bound, cubical filtration, margin test, audit, coverage, certified pruning objective	unchanged in form
GT topology from a rendered mask	GT component count per class from boxes; synthetic testbed uses renderer counts
$\varepsilon_{\text{cert}}$ product over the decoder	closed form over the final class conv only (Equation (2)); wider scope needs IBP

- Violations $> 0 \Rightarrow$ the bound, the filtration convention, or the implementation is wrong. Find it. *Do not tune τ to make violations disappear.*

We also report *certified coverage*: the fraction of triples that certify at each budget. A sound certificate that covers 4% of triples is honest but weak, so the coverage-versus-budget curve is a required figure.

3 Implementation

3.1 The retarget, precisely

Table 1 lists what transfers verbatim from the plan and what changes. The YOLO26 `Detect` head computes, per level i , `scores = cls_head[i](x[i])` with shape $[B, n_c, H_i, W_i]$ and applies `.sigmoid()` only later; the hook output is therefore exactly the logit map. For an `end2end` model (the YOLO26 default) the deployed predictions come from the `one2one` head, so we hook `one2one_cv3[i]`.

3.2 Package

The `topo_prune/` subtree is self-contained (the same pattern as the repository’s earlier `ood_monitor/` workstream) and leaves the parent training code untouched.

`filtrations.py`

superlevel cubical PH on a 2-D logit map (GUDHI); the linear-time Euler-characteristic curve as the cheap control. Verified against hand-computed diagrams for a disk, an annulus, and two disjoint disks.

`certificate.py`

margin, certify, the falsification audit, coverage.

`metrics.py`

Betti-number error (the primary metric); a simplified 2-D Betti-matching error.

`head_maps.py`

hook `one2one_cv3[i]`, return the per-scale $[n_c, H, W]$ logit maps.

prune.py

structured magnitude/random pruning of the final class-conv input channels.

lipschitz.py

ε_{emp} , per-layer spectral norms, the closed-form $\varepsilon_{\text{cert}}$.

synthetic.py

logit-map scenes with exact ground-truth homology.

report.py

run manifests $\rightarrow$ Markdown tables (no hand-entered numbers).

3.3 TDA backend and the Euler control

We use GUDHI, not Cubical Ripser: the parent repository already standardised on its `CubicalComplex`, it installs on the workstation’s Python 3.14, and re-using the backend keeps the new diagrams comparable to the repository’s existing topology numbers. The Euler-characteristic curve is computed directly as the alternating sum of k -cell counts across the filtration sweep (vectorised, genuinely $O(\text{pixels} + \text{thresholds})$), so it is reproducible without a differentiable-ECT library.

3.4 Pruning scope

We prune the **input channels of the final 1×1 classification convolution** `one2one_cv3[i][-1]`. Because that convolution is linear, a dropped input channel’s contribution to the logit map is removed exactly, which keeps the certified propagation path a single layer and makes Equation (2) exact. The producing convolution and its normalisation are zeroed as well so the FLOP saving is physical, but this does not change the logit map (the channel is already gone at the final conv). Selection is by $\|W_{\text{final}}[:,k]\|_2$ (magnitude) or a seeded random subset. The topological selection criterion is deferred to P2. Full dependency-aware removal across the neck is also a P2 concern and matches the parent repository, whose dependency engine is marked *pending checkpoint integration*.

4 Experimental setup

Environment. Windows, Python 3.14.5, no CUDA GPU. The topological core (GUDHI 3.13, SciPy 1.18, NumPy 2.5) and the detector stack (PyTorch 2.14 CPU, Ultralytics 8.4.9 from the pinned `third_party/ultralytics` submodule) all install and run. CPU inference of YOLO26-nano at 640 px is ≈ 50 ms per image.

Stand-in checkpoint. The WSD export and the tuned spacecraft checkpoint are not on any reachable host, so **COCO-pretrained yolo26n.pt** ($n_c = 80$) stands in and COCO128 is the image set. The pipeline is checkpoint-agnostic; every result below is expected to be re-run on the WSD checkpoint when it lands, which is when the numbers become domain-meaningful.

Splits. COCO128’s 128 images are split 64 calibration / 64 held-out test. ε_{emp} and $\varepsilon_{\text{cert}}$ are computed on calibration only; the audit runs on test only. Operating point $p = 0.25$, so $\tau = \text{logit}(0.25) = -1.0986$. A class is audited on an image only if its logit map peaks above $\text{logit}(0.05)$ somewhere (otherwise $\beta = 0$ trivially on both sides). This yields 555 audited (x, c, i) triples per arm.

Table 2: E1. COCO-pretrained YOLO26-nano, 640px, $\tau = \text{logit}(0.25)$, 64 calib / 64 test, 555 audited triples per arm. “cov” = certified coverage; “ $\Delta\beta$ ” = triples whose Betti number actually changed under pruning; “viol” = audited certificate violations; “held” = ε_{emp} upper-bounded every test perturbation it was applied to. ϵ values are per- (c, i) medians, in logit units.

method	budget	ε_{emp}	$\varepsilon_{\text{cert}}$	loose.	$\text{cov}_{\varepsilon_{\text{emp}}}$	$\text{cov}_{\varepsilon_{\text{cert}}}$	$\Delta\beta$	$\text{viol}_{\varepsilon_{\text{emp}}}$	$\text{viol}_{\varepsilon_{\text{cert}}}$	held
magnitude	10 %	1.99	5.01	$2.7\times$	55.3 %	27.9 %	40	0	0	no
magnitude	20 %	3.19	10.71	$3.3\times$	41.3 %	4.0 %	63	0	0	no
magnitude	30 %	4.38	17.07	$4.5\times$	33.2 %	0.4 %	80	0	0	no
magnitude	40 %	5.29	23.12	$5.0\times$	26.8 %	0.0 %	100	1	0	no
magnitude	50 %	6.30	29.46	$5.1\times$	20.4 %	0.0 %	127	1	0	no
random	10 %	3.84	9.18	$2.3\times$	36.8 %	13.2 %	50	0	0	no
random	20 %	5.09	14.56	$1.8\times$	26.3 %	1.4 %	85	1	0	no
random	30 %	10.79	29.83	$3.7\times$	16.6 %	0.0 %	104	0	0	no
random	40 %	70.06	107.28	$4.5\times$	4.3 %	0.0 %	138	0	0	no
random	50 %	68.50	111.75	$3.7\times$	0.0 %	0.0 %	138	0	0	yes

5 E1: falsification audit on a really-pruned YOLO26-nano

We prune the classification head at target budgets $\{10, 20, 30, 40, 50\}\%$ of its channels, by two methods (**magnitude**, **random**) — ten arms. For each arm we capture reference and pruned logit maps on both splits, compute ε_{emp} and $\varepsilon_{\text{cert}}$ per (c, i) on calibration, and audit all 555 test triples. Table 2 is generated verbatim from the run manifest.

5.1 Findings

The sound certificate never fails its audit. Across all ten arms — 5,550 triples, of which 891 certify under $\varepsilon_{\text{cert}}$ — the $\text{viol}_{\varepsilon_{\text{cert}}}$ column is **zero**. This is the P1 falsifiable prediction, and it holds. The certificate implementation, the superlevel/negation convention, and the diagram-threshold bookkeeping are internally consistent.

Certified coverage collapses with budget. Under $\varepsilon_{\text{cert}}$, coverage is 27.9 % at a 10 % magnitude budget, 4.0 % at 20 %, and effectively zero by 40 %. The median $\varepsilon_{\text{cert}}$ grows roughly linearly ($5.0 \rightarrow 10.7 \rightarrow 17.1 \rightarrow 23.1 \rightarrow 29.5$ logits), while the margins of the reference diagrams do not, so fewer and fewer triples clear $m > \varepsilon_{\text{cert}}$. This is precisely the “honest but weak” regime the plan anticipates: a real guarantee, on a shrinking fraction of inputs. The coverage-versus-budget curve is the headline figure and it is unflattering — which is the correct scientific outcome to report.

The empirical bound is genuinely unsound out of distribution. On 9 of 10 arms the calibration ε_{emp} failed to upper-bound at least one test-split perturbation (**held** = no). The three $\text{viol}_{\varepsilon_{\text{emp}}}$ entries are exactly such cases: on image 000000000443.jpg at class 65, level 2, the true sup-norm gap was 6.10 and 7.35 against an ε_{emp} of 4.42 and 4.38; on 000000000502.jpg at class 21, level 2, the gap was 5.88 against 3.33. In every case `epsilon_bounded_the_gap` is **false** in the manifest. These are not certificate bugs; they are the reason ε_{emp} is a diagnostic, not a certificate. (The lone **held** = yes row, **random** 50 %, has an ε_{emp} so large — 68 logits — that even the out-of-distribution test gaps on the specific audited classes stayed under it.)

Table 3: E2. 240 synthetic scenes, 72×72 grid, 1,200 perturbed cases per δ . Constructed-versus-computed Betti mismatches over all 240 scenes: **0**.

δ ($\ \text{perturbation}\ _\infty$)	audited	certified	coverage	cases that changed Betti
0.5	1200	1200	100.0 %	0
1.0	1200	1200	100.0 %	0
1.5	1200	1200	100.0 %	0
2.0	1200	1070	89.2 %	25
2.5	1200	955	79.6 %	101
3.0	1200	795	66.2 %	202
4.0	1200	430	35.8 %	412
5.0	1200	175	14.6 %	655
6.0	1200	35	2.9 %	890

Looseness. The certified bound is $1.8\text{--}5.1\times$ the empirical one, growing with budget (more dropped channels, more slack accumulated in the sum of Equation (2)). A single-layer bound at $\approx 3\text{--}5\times$ looseness is far better than the “astronomically large” product of spectral norms the plan feared for an encoder–decoder path; scoping the certificate to the head is what buys this.

The metric is sensitive. The number of triples whose Betti number *actually* changed under pruning rises monotonically with budget, from 40 (10 % magnitude) to 127 (50 %) and 138 (random $\geq 40\%$). Pruning really does change output topology, and the Betti-number error registers it.

6 E2: synthetic validation with exact ground-truth homology

E1 cannot check soundness against *truth* — only against the pruned model’s own diagram. E2 does. We construct 240 scalar fields standing in for logit maps, whose superlevel-set Betti numbers at the scene threshold are exact by construction: $n \in \{1, 2, 3, 4\}$ bright vertical panels with a controllable gap ($\beta_0 = n$ open, 1 closed) and an optional disjoint annulus ($\beta_1 += 1$). Per-scene contrast is randomised so the margin spreads. Each scene is then perturbed the way pruning perturbs a logit map — a smooth low-frequency field rescaled to an exact target sup-norm δ — for five seeds per δ , and audit is run. Table 3 is verbatim from the manifest.

6.1 Findings

Zero mismatches, zero violations. Every one of the 240 scenes’ constructed Betti numbers equals its computed superlevel-set Betti numbers — the filtration and the reader agree with ground truth. Across all $9 \times 1200 = 10,800$ audited perturbed cases the violation count is **zero**. When the certificate says “preserved,” the Betti numbers are in fact preserved, against the true homology and not merely against the model’s own recomputation.

Coverage decays smoothly and correctly. For $\delta \leq 1.5$ every scene’s margin exceeds δ and coverage is 100 % with no Betti changes anywhere. As δ passes the spread of scene margins, coverage falls ($89.2\% \rightarrow 2.9\%$) in lock-step with the fraction of perturbations that break topology ($25 \rightarrow 890$ of 1200). Crucially, at every δ the certificate declined *all* the cases it needed to: none of the Betti-changing perturbations were certified.

Table 4: Option B, COCO128 pipeline check, `grabcut`. “changed” = the mask’s total Betti numbers differ from the box-fill mask’s.

images (/ with boxes)	128 / 126
images where GrabCut Betti $\neq$ box-fill Betti	105 (83.3%)
total β_1 introduced vs box-fill	719
total $ \Delta\beta_0 $ vs box-fill	1,372
images where GrabCut fell back on ≥ 1 box	73
mean mask coverage of the image	29.4%
top class by images changed (person, id 0)	40/61, $\Sigma\beta_1 = 210$

Targeted adversarial check. For each of the 165 multi-component scenes we added a horizontal bright band that deliberately merges the panels (β_0 drops). The band changed the Betti numbers on 90 of them (the others were already single-component or the band missed the panel band under the ring layout). The certificate certified the merge as topology-preserving on 0 of 165.

7 Option B: segmentation masks from bounding boxes

A parallel branch pursues the cleaner route: if the detector’s boxes can be turned into component *masks*, the plan’s certificate applies verbatim to a single segmentation logit map, with no per-scale reinterpretation. The open question — and the reason it is a branch, not the main line — is whether such masks carry topology the boxes do not. A box-fill mask is a solid rectangle: β_0 equals the box count and β_1 is identically zero, i.e. no information beyond the detector’s own instance count.

7.1 Method

Three mask sources of increasing cost: `boxfill` (the degenerate baseline and the fallback); `grabcut` (OpenCV GrabCut, one pass per box on a padded crop, no training or GPU, effective when the box border is background); and `sam` (box-prompted Segment Anything, lazy, optional weights). Overlaps resolve smaller-box-wins. `topology_check` computes β_0 (connected components) and β_1 (Euler-number identity on the padded complement) of a binary mask via SciPy, verified against a disk, an annulus, and two disks.

7.2 Pipeline check on a COCO stand-in

With no WSD export, the mechanism was exercised on COCO128 (YOLO boxes). Table 4 summarises. GrabCut masks differ topologically from box-fill on 105 of 126 box-carrying images (83.3%), introducing 719 holes (β_1) in total against box-fill’s zero. The measurement pipeline produces a real, non-degenerate signal.

Caveat. GrabCut fell back to a rectangle for at least one box on 73 of 128 COCO images: COCO’s cluttered, overlapping-box scenes are the hard case. WSD — one spacecraft per image, dark background, well-separated components — should fall back far less, so 83% is not the WSD gate number. The gate (proceed to train a segmentation head iff the masks change the topology on a meaningful fraction of images, concentrated in the elongated classes) needs the WSD export.

8 Threats to validity

1. **COCO, not WSD.** Every number here is on a COCO-pretrained detector and COCO images. The certificate machinery is domain-independent, but “the pruned spacecraft model provably preserves component topology on $Y\%$ of frames” cannot be stated until the run is repeated on the WSD checkpoint. The $n_c = 80$ COCO head is also wider than WSD’s $n_c = 4$, which inflates the audited-triple count and the class-conditional structure.
2. $\varepsilon_{\text{cert}}$ **is calibration-based.** M_k in Equation (2) is a maximum over the calibration set, not over the input domain. A test image with a larger penultimate activation than any calibration image can, in principle, violate $\varepsilon_{\text{cert}}$ too — it did not here, absorbed by the $1.8\text{--}5.1\times$ looseness, but the guarantee is “high-confidence” in the plan’s sense, not worst-case.
3. **Single-layer pruning scope.** The clean closed-form bound holds only because pruning is confined to the final classification convolution’s inputs. Widening the scope (into the neck) reintroduces the multi-layer Lipschitz product and its looseness.
4. **Per-level certification is conjunctive and untuned.** We certify each (c, i) independently; an image is fully certified only if all its (c, i) certify. Whether the assigned-level-only variant (certify the level YOLO assigns each GT object to) is materially stronger is untested.
5. **Synthetic scenes are simple.** E2’s scenes are axis-aligned panels and circular annuli with sub-threshold texture. They exercise β_0 merges and β_1 fills but not, e.g., thin filaments or near-tangent components where the cubical discretisation is fragile.
6. **No retraining.** All E1 numbers are before any fine-tuning. Retraining is a confound that can rescue a bad mask; the pre-retraining gap is the cleaner causal signal, but it is also the pessimistic one for coverage.
7. **Betti-matching error is a simplified 2-D approximation.** The full induced-matching metric of Stucki et al. needs the spatial location of each critical cell, which the raw diagram does not carry; our version optimally matches alive-at- τ bars on endpoint distance and counts the residuals.

9 Future work

9.1 A worst-case certified bound via interval bound propagation

The single most valuable next step. Replace M_k in Equation (2) — a calibration maximum — with a sound upper bound on $|\phi_k^{(i)}|$ over the input domain (pixels in $[0, 1]$, or an ℓ_∞ ball around the calibration data). Interval bound propagation through the backbone, neck, and head prefix gives exactly this: propagate $[\underline{x}, \bar{x}]$ forward through each convolution (split weights into positive and negative parts), each BatchNorm (affine, monotone), and each SiLU (bounded, with known monotonic envelope). The result is a $\varepsilon_{\text{cert}}$ that is a certificate in the strict sense, at the cost of additional looseness. Two sub-questions: (a) how much looseness does IBP add on top of the $3\text{--}5\times$ we already see, and does CROWN-style linear relaxation recover enough of it to keep coverage non-trivial; (b) can the IBP pass be restricted to the receptive field of each output cell, since the cubical margin is a local quantity.

9.2 Widening the pruning scope

Move the prunable unit from the final class-conv inputs into the last one or two neck blocks (the C3k2 blocks feeding the head). The certified path is then 2–4 convolutions and the Lipschitz product no longer collapses; we would measure whether the empirical looseness stays under $\sim 10\times$ and whether a dependency-aware structural removal (coupling normalisation with its producer, propagating indices through concatenations) keeps the FLOP saving physical. This also connects to the parent repository’s pending dependency engine.

9.3 The certified pruning objective (P2/P3)

Once $\varepsilon_{\text{cert}}$ is a function of the mask, pruning stops being a ranking problem:

$$\min_M \varepsilon_{\text{cert}}(M) \quad \text{s.t.} \quad \text{FLOPs}(M) \leq B.$$

Greedy first (drop the channel with the smallest marginal contribution to the bound, recompute, repeat); a small ILP over per-layer budgets only if greedy is visibly loose. This is the `cert-min` arm of P2 and the criterion the whole project argues for: prune to minimise a certified bound on barcode displacement, not a proxy score.

9.4 The two-sided intervention (P2)

The experiment that separates this work from the correlational literature. At each budget, construct FLOP-, channel-count-, and ℓ_1 -mass-matched masks: `topo-low` (prune what the topological score calls redundant), `topo-high` (prune what it calls important — the necessity arm), `random`, `l1-low`, `taylor`, `ecc-low`, `cert-min`. Five seeds each; report before- and after-short-retraining separately. Primary metric: Betti-number error, not mAP. Pre-registered hypotheses: at matched FLOPs, `topo-high` degrades Betti error significantly more than `topo-low`, and the gap exceeds `random-topo-low`; `cert-min` beats `l1-low` and `taylor` at equal FLOPs before retraining. The kill criterion is real: if `topo-high` and `topo-low` are statistically indistinguishable, the score is not difference-making and that is a publishable negative result about a class of methods.

9.5 The Euler-characteristic control

Run the entire pipeline with homology replaced by the linear-time Euler-characteristic curve (already implemented, `descriptors.ecc_vector`). If the ECC-based selection matches the homology-based one within noise, that is a *better* result for edge deployment, not a failure: a $\sim 100\times$ cheaper descriptor suffices. This is a mandatory control, not an optional one.

9.6 P0 hygiene, retrofitted

The variance/permutation ablations (P0.2–P0.4) were specified for the parent repository’s PEEK-map score and have not been run against the retargeted logit-map construction. They should be: regress the per-channel topological importance on per-layer weight variance, weight ℓ_1 mass, activation Frobenius norm, and effective rank; permute weights within each layer and recompute; apply topology-preserving vs. topology-changing augmentations and check the score tracks the latter. If weight variance alone explains $> 85\%$ of the score’s variation, the criterion is not measuring topology and must be redesigned before P2.

9.7 Topological restoration (P4)

After pruning, add a loss term pulling the pruned logit barcode back toward the unpruned model’s, using a differentiable persistence layer or the Betti-matching loss. Two purposes: causal evidence (if restoring the barcode restores accuracy while a matched-magnitude geometric loss does not, topology is difference-making) and a shipping method (topology enters training, is absent from inference — never violating the no-persistent-homology-on-the-inference-path rule).

9.8 Multi-scale certification strategy

Decide empirically between (a) independent per- (c, i) certification with a conjunctive image-level flag (current), (b) certifying only the level YOLO’s assignment owns each GT object at, and (c) a fused single-grid map via upsampling, which introduces an interpolation Lipschitz factor into Equation (1). Measure the coverage each yields on the WSD checkpoint.

9.9 Zigzag persistence as a stopping rule (P5, stretch)

Treat progressive pruning as its own filtration — channels appear/disappear along the pruning trajectory — and read off when the last long-lived output feature dies. That death is a principled candidate for the pruning-ratio stopping point, replacing a hand-set budget.

9.10 Finishing Option B

If the WSD feasibility gate passes: train a small segmentation head (or a UNet) on the Grab-Cut/SAM masks, and run the certificate on its single logit map — recovering the plan’s original, cleaner formulation. Compare its certified coverage at matched FLOPs against the detection retarget. If the gate fails (GrabCut masks are box-fill with noise), record it and stay on the detection line.

9.11 On-device measurement (P3)

Export the certified-pruned model to ONNX/TensorRT, quantise to int8, and measure on the Jetson Orin Nano: median and tail latency, and — reported as first-class, because on-board compute is memory-bound — peak activation memory. The headline deployment figure is Betti-error versus peak memory, one curve per pruning arm, not mAP versus FLOPs.

10 Conclusion

P0 found that the plan’s segmentation certificate does not fit the repository’s detector, and P1 shows the retarget is sound: the classification logit maps of YOLO26’s `Detect` head have real superlevel-set homology, the bottleneck stability bound transfers unchanged, and the falsification audit passes — zero violations across 5,550 real triples (E1) and 10,800 synthetic ones with exact ground truth (E2). The honest limitation is coverage: a single-layer, calibration-based certified bound guarantees topology preservation on 27.9% of triples at a 10% head-prune budget and almost none by 40%. Making that number larger — a worst-case bound via IBP, a wider but still-short certified path, and the `cert-min` pruning objective — is the substance of the next phase, alongside the P2 two-sided intervention that will decide whether the topological criterion is difference-making at all.

A Reproduction

From the repository root, on a machine with Python 3.10+ (3.14 works):

```
python -m venv .venv && . .venv/Scripts/activate
pip install gudhi scipy numpy pyyaml joblib pytest
pip install --index-url https://download.pytorch.org/whl/cpu \
    torch torchvision
pip install -e third_party/ultralitics

# 53 tests, detector-free core + prune (skips without torch)
make -C topo_prune test
# E1: capture, prune, audit
python topo_prune/experiments/p1_certificate/run_p1.py
# E2: synthetic validation
python topo_prune/experiments/p1_certificate/synthetic_p15.py
# regenerate docs/p1-results.md from the manifests
python topo_prune/src/topo_prune/report.py
```

Run manifests are written to `topo_prune/experiments/p1_certificate/runs/`; the Markdown tables in `topo_prune/docs/p1-results.md` and every number in Tables 2 and 3 are generated from them.

B Key definitions and conventions

Superlevel diagram

persistence of the filtration $\{f \geq t\}$ as t decreases; stored in f -value units with birth $\geq$ death, essential death $= -\infty$.

Betti at τ

$$\beta_k(\{f \geq \tau\}) = \#\{(b, d) \in D_k : b \geq \tau > d\}.$$

Operating point

$\tau = \text{logit}(p)$; this report uses $p = 0.25$.

Audited triple

a (test image, class, detection level) for which the class clears $\text{logit}(0.05)$ somewhere on that image.

Certified coverage

fraction of audited triples with $m(\tau, D) > \varepsilon$.

Violation

a certified triple whose pruned-model Betti numbers differ from the reference; expected count 0.